Online Vegetable Management System and Crop Prediction System

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Abstract

Crop selection, market volatility, and restricted access to digital trading platforms are common issues in developing countries' agriculture. In order to give farmers better cultivation planning and direct market access, this project suggests an intelligent, farmer-centered e-commerce and decision support system that makes use of machine learning and mobile technology. A machine learning module created with XGBoost serves as the system's first essential component. It has a regression model to forecast the anticipated market price at harvest time and a classification model to suggest the top five most appropriate crops based on cultivation and harvesting dates. To assist farmers in making well-informed decisions prior to planting, these forecasts are produced using historical agricultural records saved in CSV format as well as current weather data. A user-friendly mobile application that connects farmers and consumers via a digital marketplace is also integrated into the system. Through the app, farmers can easily post their harvested produce, and consumers can easily browse and buy the products that are available. By doing away with middlemen, this feature guarantees fair pricing and boosts farmers' profitability. The suggested system encourages digital transformation in rural farming communities and supports sustainable agriculture by fusing predictive analytics with an online trading platform. Stronger ties between farmers and buyers, less waste, and smarter farming are all benefits of this all-encompassing strategy.

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Acronyms and Abbreviations

API	Application Programming Interface
CSV	Comma-Separated Values
DSS	Decision Support System
E2E	End-to-End
FB	Farmer and Buyer
GUI	Graphical User Interface
ML	Machine Learning
RT	Real-Time
UI	User Interface
XGBoost	Extreme Gradient Boosting
MLM	Machine Learning Model
RF	Rainfall
TEMP	Temperature
HUM	Humidity
VSMS	Vegetable Stock Management System

Chapter 1

"Instantaneous" Online vegetable stock management system and crop Prediction syste

1.1 Scope

Presenting the findings of the authors' final year research project, "Crop and Price Prediction System with Integrated Farmer–Buyer Marketplace", which they developed during the academic year 2024/2025, is the focus of this thesis. A machine learning-based decision support system that helps farmers with crop planning and price forecasting is being designed and put into use as part of this project. A user-friendly mobile application that facilitates direct trade between farmers and consumers is also integrated into the system. The project presents a useful e-commerce platform for agriculture and integrates real-time weather data with classification and regression techniques.

1.2 Background and Motivation

Many nations' economies, including Sri Lanka's, rely heavily on agriculture. But farmers frequently struggle to choose which crops to grow and to get timely access to markets. Conventional farming practices primarily rely on experience rather than data, which increases financial risk and wastes resources.

- Using historical data and current weather, make data-driven crop recommendations.
- Utilize regression techniques to forecast market prices for specific crops at harvest time.
- Give farmers the information they need to make informed decisions before planting.
- By providing a digital marketplace via a mobile application, you can close the gap between farmers and this system seeks to improve agricultural productivity, lower risk, and provide farmers and consumers with cutting-edge digital tools by fusing predictive analytics with e-commerce.

1.3 “Instantaneous” Difficulties with Crop Planning and Market Access

Crop selection and sales planning are often done in farming communities without access to accurate weather forecasts or market data. Financial losses frequently follow from low market prices, mismatched crop types, and poor yield.

1.3.1 Requirements

The following essential needs are met by the project’s suggested solution:.

- Reliable crop recommendations: Determined by historical performance, current weather conditions, and the dates of cultivation and harvest.

- Forecasting market prices: Applying a regression model that has been trained using past price data to information to forecast anticipated harvest prices.

- Low computational cost: The system needs to be small enough to operate on web or mobile platforms that are within rural farmers’ reach..
  -to-use mobile app: A straightforward, multilingual interface that allows buyers and farmers to post goods and place orders.

- Integration and real-time updates: Constant access to current product listings and weather APIs for efficient user interaction.

- Scalability and security: The ability to scale across various regions or crop types while maintaining secure data handling.

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Chapter 2

Foundations and Needs of the System

This chapter describes the machine learning solution architecture using XGBoost, the use of real-time weather data via an API, and the agricultural challenges that the suggested system attempts to address. To help farmers make decisions about crop selection and market planning, the system combines predictive analytics with an online shopping platform.

2.1 Using Machine Learning to Predict Prices and Crops

XGBoost (Extreme Gradient Boosting), a potent and scalable tree-based ensemble algorithm, was used to build the two main machine learning models that power the system. These models provide real time recommendations based on current weather and updated agricultural data

2.1.1 Model of Crop Classification

,Based on the following characteristics, the classification model suggests the top 5 most suitable crops for a farmer: we need to give input as Cultivation Date,Famer Location and harvesting Date indeed and we get live weather details using weather API to display in crop prediction system interface and train ML Model to predict suitable vegetable price before cultivating in harvesting periode.

Chapter 3

Prototype Design

3.1 Implementation of a Prototype System

3.1.1 Implementation

Given the advantage of a series-hybrid configuration in Fig.3.1, the first implementation and related tests were carried out based on the simplified block diagram of Fig.3.2.

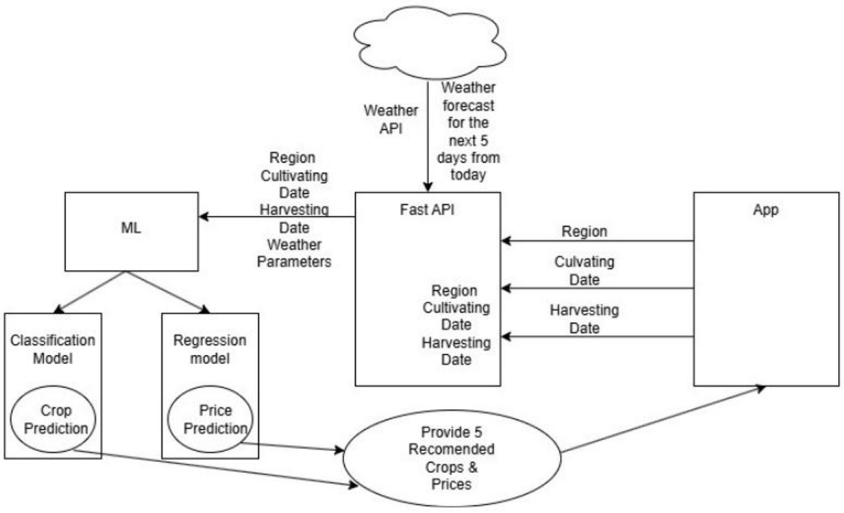


Figure 3.1: System Architecture

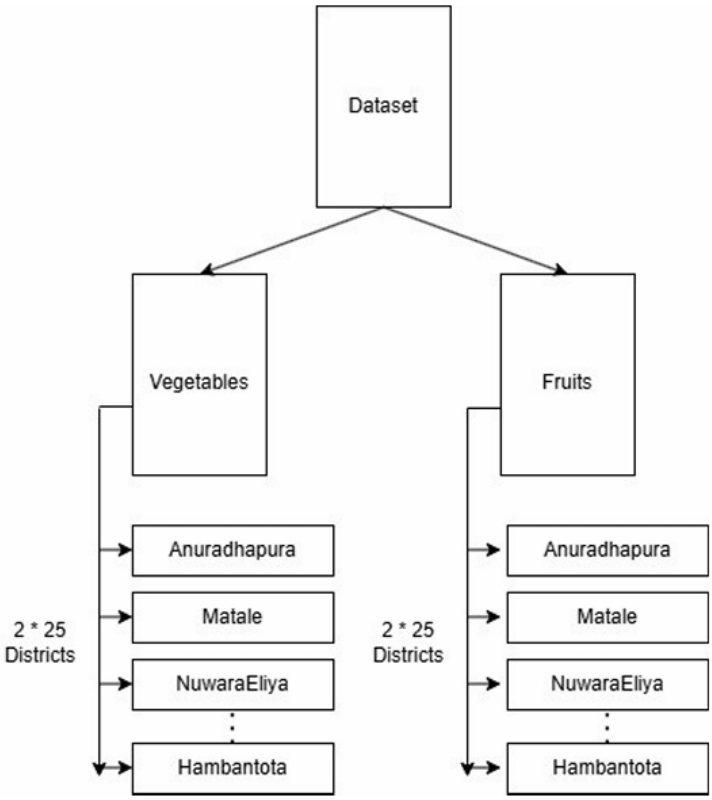


Figure 3.2: Data Classification

Chapter 4

Real-Time Prediction and Stock Management Requirements

4.1 Fast Prediction Requirement in the Crop Advisory System

This chapter discusses the system's performance requirements for providing real-time crop and price predictions using machine learning, as well as instant responsiveness in the integrated vegetable stock management module. The system must operate reliably on mobile devices and rural networks, ensuring fast and accurate recommendations and smooth trading between farmers and buyers.

4.1.1 Fast Prediction Requirement in the Crop Advisory System

As described in earlier chapters, the system provides crop recommendations and market price forecasts using machine learning models built with XGBoost These predictions rely on a combination of user inputs,historical datasets, and daily updated weather parameters from a Weather AP To ensure usability in farming conditions, predictions must be delivered in under 2–3 seconds, even with limited processing power and variable internet availability. This requirement includes: Fetching current weather data Extracting features Running classification and regression models Displaying results to the user

4.1.2

4.2 Real-Time Stock Management Requirement

In addition to predictions, the system features a Vegetable Stock Management Sytem built into the mobile application. This e-commerce module allows farmers to post their harvested produce, and customers to browse and purchase items in real-time.

4.2.1 Stock Posting and Buying Flow

The user flow involves:

1. Farmer posts product details:Crop name, quantity, price, harvest date, expiry

2.The system stores the product in a real-time database (for example, Firebase).

3.Buyer sees product listing instantly and places an order

4.The system updates the stock quantity and order history.

This process must occur without delay. Any delay in stock updates could lead to double-selling, expired listings, or customer dissatisfaction.

To meet these demands, the system uses:

Firebase real-time database for product listing and ordering Live listeners to update the UI immediately when product stock changes Validation logic to prevent posting expired or unavailable products

4.1.3 Data Synchronization Challenges

Real-time performance must be maintained even when:

* Internet is slow or intermittent * API response delays occur * Multiple users interact with the same product

To ensure stability:

Weather API responses live weather in cultivation date. Stock updates areatomic, ensuring consistent product quantities Mobile app can store offline data and sync when reconnected

Chapter 5

XGBoost Regression and Classification and Objectives

5.0.1 XG Boost Regression Objective

$$Obj = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \sum_{k=1}^{K}\Omega(f_k) \tag{5.1}$$

where:

- y_i: actual price
- $\hat{y}_i$: predicted pri
- $\Omega(f_k)$: regularization term defined by:

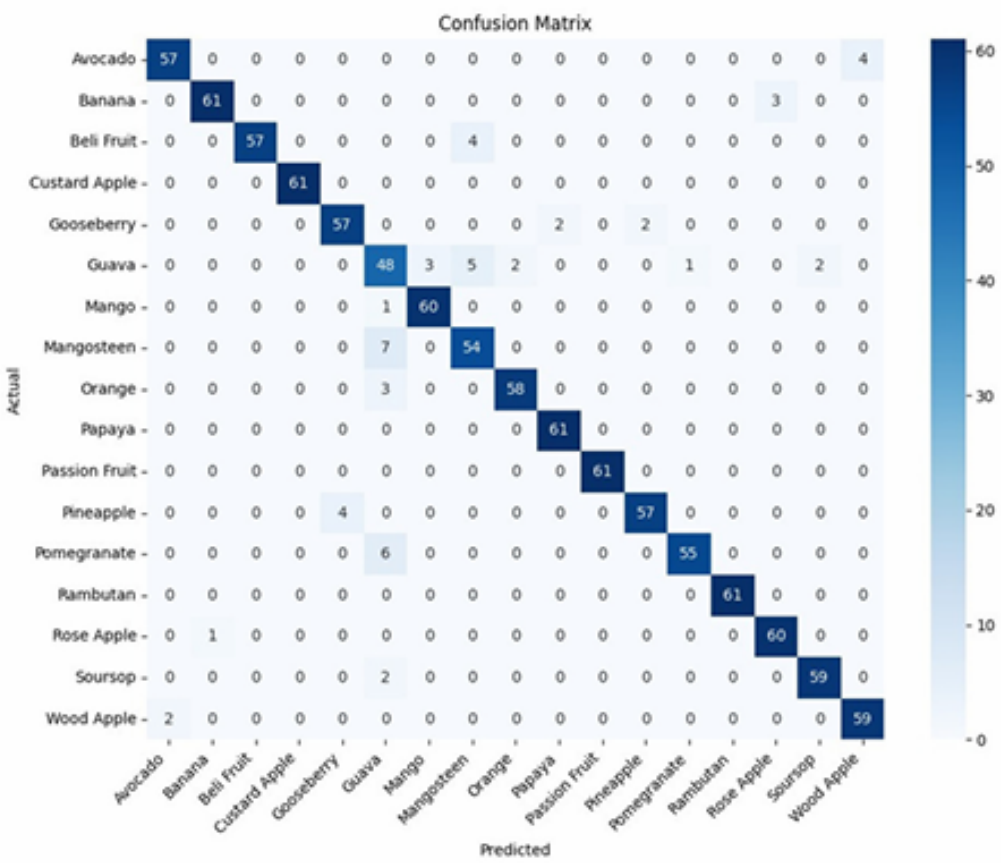
$$\Omega(f_k) = \gamma T + \frac{1}{2}\lambda\|w\|^2 \tag{5.2}$$

with T: number of leaves, w: leaf weight vector, γ, λ: regularization parameters.

5.0.2 XGBoost Classification Log Loss

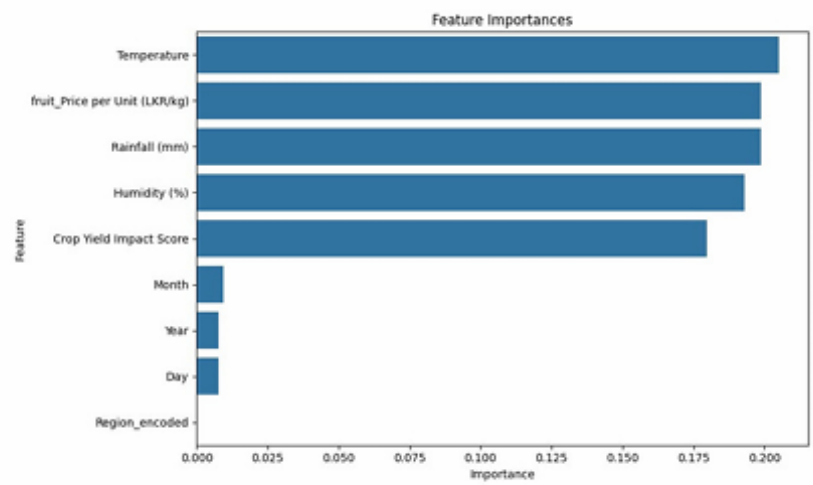
$$\text{logloss}(y_i, \hat{y}_i) = -y_i \log(\hat{y}_i) - (1 - y_i)\log(1 - \hat{y}_i) \tag{5.3}$$

This penalizes incorrect prediction,improving crop recommendation accuracy.



Confusion Matrix of the Matale Fruit

Figure 5.1: Feature importance of the regression



Feature importance of the Regression

Figure 5.2: Feature importance of the regression

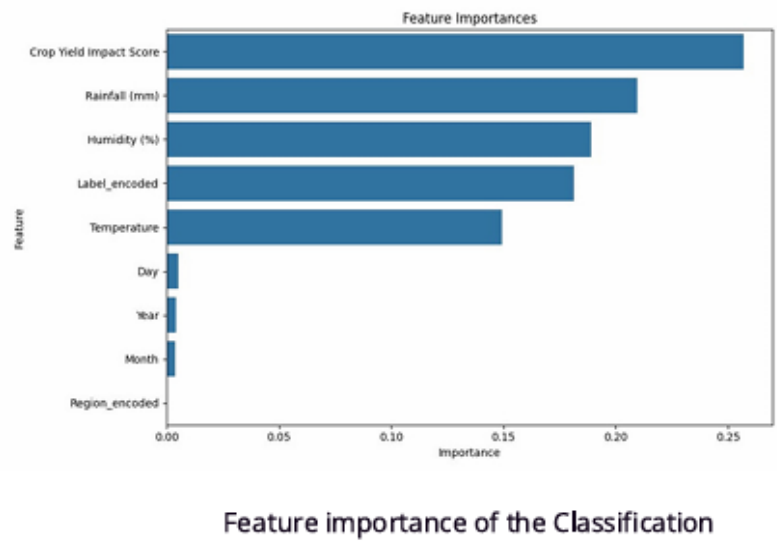


Figure 5.3: Feature importance of the classification

Chapter 6

Conclusions and Future Work

6.1 System Deployment

The system is to be deployed as a complete mobile and web application. It will provide a user-friendly experience for farmers and consumers. Using cross-platform development frameworks such as Flutter, the application will be able to:

- Provide real-time crop recommendations
- Live market price forecasting
- Stock control and tracking of sales
- Multilingualism (Sinhala and Tamil planned)

6.2 Planned Enhancements

Planned future developments include:

- Live Weather Integration: Dynamic crop recommendation updates with real-time weather API.
- Blockchain-Based Price Logs: Immutable price record storage to ensure data authenticity.
- Expert Recommendations: Leverage agronomist domain expertise for en- hanced decision support.
- Advanced Notifications: Market trend, expiry date, and buyer request notifi- cations.

6.3 Scalability and Outreach The system will be extended to multiple agricultural zones across Sri Lanka, with potential regional customization based on localized datasets. Long-term opportu- nities in-

- Integration with national agricultural advisory system
- Partnerships with local cooperatives
- Offline sync and low-bandwidth mode of operation 6.3 Scalability and Outreach 7

All these features should significantly increase adoption by rural farmers and promote smarter farming.

6.3 Future Work

Battery Storage Optimisation: o The battery storage configuration requires more investigation.
Supercapacitor Optimisation: o The current system is showing that a number of the specifications are being reached, or are close to being reached. Additional supercapacitors will be introduced to the system, to hit all targets (including worst case scenarios).

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Chapter 7

Appendix A: Publication

Appendix A lists a series of publications developed as part of the project, such as conference papers and journal submissions. All the contributions enumerated below are original work that was carried out by the authors as part of the final year research project.

A 1. K.P.G.H.D.Dissanayake, J.M.Widanegamage , "A Machine Learning Based Decision Support System for Crop Recommendation and Price Prediction," in *Proc. of the International Conference on Smart Agriculture and Rural Development*, Colombo, Sri Lanka, June 2025.

A 2. K.P.G.H.D.Dissanayake and J.M.Widanegamage, "Design of a Farmer–Buyer Integrated Mobile E-Commerce Platform using Flutter and FastAPI," *Journal of Agricultural Informatics and Technology*, submitted for review, July 2025.

A 3. K.P.G.H.D.DissanayakeN, "Crop Selection Using XGBoost Classification with Real-Time Weather Integration," Poster presentation at *CSNE Undergraduate Research Symposium*, University of Sri Lanka, May 2025.

A 4. K.P.G.H.D.Dissanayake, J.M.Widanegamage"Digitally Empowering Farmers in Sri Lanka: An AI-Based Approach," white paper submitted to *Ministry of Agriculture - Innovation Division*, April 2025.